SWRT for nonlinear four-wave interactions in ecWAM

SWRT (sparse Webb–Resio–Tracy) computes the nonlinear four-wave interactions S_{nl} from K configurations of the WRT quadrature table, each a quadruplet and its mirror image, with one optimized coefficient per configuration. SWRT is implemented in ecWAM's `SNONLIN` as an option next to the Discrete Interaction Approximation (DIA). Two files of configurations are provided for each grid: SWRT with 31 configurations, and SWRT Lite with 7, which is faster.

The SWRT files and this guide are on the branch `swrt` of Jialunx/ecwam, a fork of ecmwf-ifs/ecwam (ecWAM 1.5.14).

1. Changes to ecWAM

The patch adds one source file and changes eight others. The default `INLSCHEME = 0` leaves ecWAM unchanged.

File	Change
`src/ecwam/snonlin_swrt.F90` (new)	`INISNONLIN_SWRT` reads the SWRT file and checks it against the spectral grid. `SNONLIN_SWRT` takes the arguments of `SNONLIN` and adds S_{nl} to `SL` and its diagonal term to `FLD`.
`src/ecwam/yowstat.F90`	Declares the switch `INLSCHEME`.
`src/ecwam/mpuserin.F90`	Reads `INLSCHEME` in namelist `NALINE`, 0 for DIA (default) and 1 for SWRT.
`src/ecwam/userin.F90`	Writes the scheme to `wam.log` and stops for other values.
`src/ecwam/initmdl.F90`	Calls `INISNONLIN_SWRT` when `INLSCHEME = 1`.
`src/ecwam/implsch.F90`	Calls `SNONLIN_SWRT` instead of `SNONLIN` in the source-term integration when `INLSCHEME = 1`.
`src/ecwam/wdfluxes.F90`	Calls `SNONLIN_SWRT` instead of `SNONLIN` for the wave-dependent fluxes at the initial time when `INLSCHEME = 1`.
`src/ecwam/CMakeLists.txt`	Adds the new file to the build.
`share/ecwam/scripts/ecwam_run_model.sh`	Passes `isnonlin`, `inlscheme` and `swrt_operator` of the yml settings to the namelist and the environment.

2. Installation

The branch can be cloned directly (video guide in the release swrt-video).

```
git clone -b swrt https://github.com/Jialunx/ecwam
```

An existing clone of ecWAM 1.5.14 takes the same changes from the patch.

```
git checkout 1.5.14
curl -LO https://raw.githubusercontent.com/Jialunx/ecwam/swrt/swrt/ecwam_swrt.patch
git apply ecwam_swrt.patch
```

ecWAM then builds as before and needs no new library. The SWRT files are in the folder `swrt/files` of the branch and go to any directory the model can read. The files for the other spectral grids of section 3 are in the same folder.

```
curl -LO https://raw.githubusercontent.com/Jialunx/ecwam/swrt/swrt/files/swrt_ec4x_K31L13.ww3op
curl -LO https://raw.githubusercontent.com/Jialunx/ecwam/swrt/swrt/files/swrt_ec_K7L7.ww3op
curl -LO https://raw.githubusercontent.com/Jialunx/ecwam/swrt/swrt/files/wrt_table_K807.ww3op
```

SWRT file	Configurations	Offsets	Use
`swrt_ec4x_K31L13.ww3op`	31	13	runs with SWRT, 3.7 times the S_{nl} cost of DIA
`swrt_ec_K7L7.ww3op`	7	7	faster runs with SWRT Lite, 1.4 times the S_{nl} cost of DIA

3. Spectral grid

Each SWRT file fits one spectral grid: the number of directions, the number NFRE of source-term frequencies and the frequency fr1 of bin ifre1 (the frequency ratio is always 1.1). 7 common grid settings for SWRT have been optimized and are

ready to use.

Directions × NFRE	First frequency (Hz)	SWRT file (K = 31)	SWRT Lite file (K = 7)
36 × 36	0.0345	`swrt_ec4x_K31L13.ww3op`	`swrt_ec_K7L7.ww3op`
24 × 36	0.0345	`swrt_d24_f0345_K31L13.ww3op`	`swrt_d24_f0345_K7L7.ww3op`
24 × 30	0.0345	`swrt_d24_n30_f0345_K31L13.ww3op`	`swrt_d24_n30_f0345_K7L7.ww3op`
24 × 40	0.0345	`swrt_d24_n40_f0345_K31L13.ww3op`	`swrt_d24_n40_f0345_K7L7.ww3op`
12 × 36	0.0418	`swrt_d12_f0418_K31L13.ww3op`	`swrt_d12_f0418_K7L7.ww3op`
24 × 36	0.0418	`swrt_d24_f0418_K31L13.ww3op`	`swrt_d24_f0418_K7L7.ww3op`
36 × 36	0.055	`swrt_d36_f055_K31L13.ww3op`	`swrt_d36_f055_K7L7.ww3op`

A run names the file of its spectral grid in `swrt_operator` (section 4). For any other spectral grid, `swrt/tools/make_swrt_files.sh` builds both files, in 16 minutes for 24 × 36. Each file is the most accurate set on synthetic training spectra that stays stable when S_{nl} alone is integrated in time.

4. Running

Setting	Where	Value
`INLSCHEME`	namelist `NALINE`	1 for SWRT, 0 for DIA (default)
`WAM_SWRT_OPERATOR`	environment	path of the SWRT file
`ISNONLIN`	namelist `NALINE`	shallow-water scaling 0, 1 or 2, as for DIA
`WAM_SWRT_DIAG`	environment	`0` switches the diagonal term off (optional)

The scripts read the same settings from three lines of the yml file.

```
physics:
    inlscheme: 1
    isnonlin: 0
    swrt_operator: /abs/path/swrt_ec4x_K31L13.ww3op
```

5. Test results

5.1 Global test

The global test runs the ecWAM test case `etopo1_era5_O48` for ten days with the settings below, once with each S_{nl}. It is a cold start: the initial spectra are JONSWAP spectra computed from the local wind.

Setting	yml	Value
Spatial grid	`grid`	O48, global, about 1.9°
Directions	`directions`	36
Frequencies	`frequencies`	29 propagated, of NFRE = 36
First frequency	`fr1`, `ifre1`	0.0345 Hz, from fr1 = 4.177248E-02 Hz at ifre1 = 3
Time step	`timestep`	900 s for propagation and source terms
Physics	`iphys`	1, ST4 (Ardhuin et al. 2010)
S_{nl}	`inlscheme`, `isnonlin`, `swrt_operator`	inlscheme = 1 with the file of WRT, SWRT or SWRT Lite, or 0 for DIA; isnonlin = 0
Period	`begin`, `end`	00 UTC 1 January 2025 to 00 UTC 11 January 2025, 10 days
Forcing	`forcings`	ERA5 10-m winds and sea ice every hour

Method	ε_{Hs} (%)	ε_{Tz} (%)	Mean H_s at 10 days (m)
WRT	–	–	2.06
SWRT	1.4	1.3	2.03
SWRT Lite	1.8	1.4	2.10
DIA	8.1	6.0	2.19

ε_{Hs} and ε_{Tz} are $\varepsilon_X = \Sigma\ |X - X_{WRT}| / \Sigma\ X_{WRT}$ for $X = H_s$ and T_z, with the sums over the hourly outputs from t = 1 h and over the sea points where H_s of WRT is at least 0.1 m.

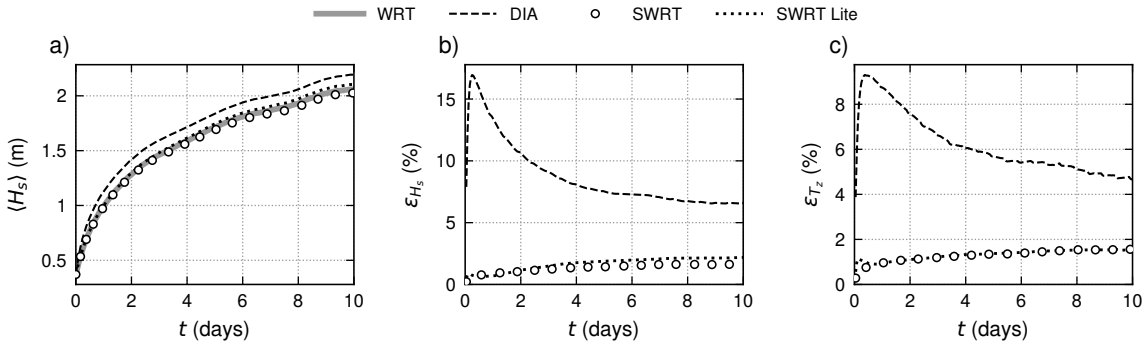


Fig. 1. Global statistics of the 10-day test on the grid of 36 × 36, with S_{nl} from WRT (thick gray line), DIA (dashed), SWRT (circles) and SWRT Lite (dotted). (a) Mean H_s over the sea points; (b) ε_{Hs} and (c) ε_{Tz} at each hour.

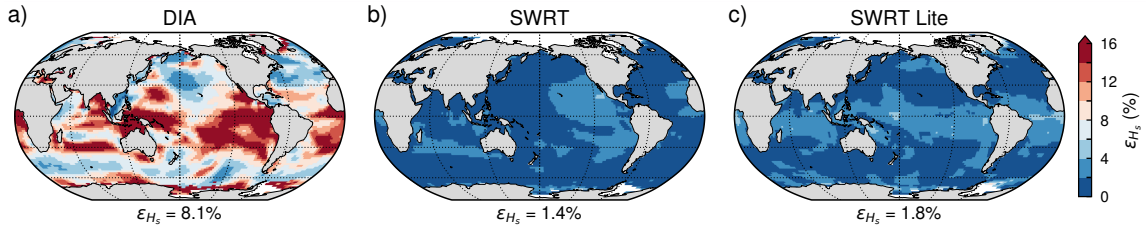


Fig. 2. ε_{Hs} at each sea point over the ten days for (a) DIA, (b) SWRT and (c) SWRT Lite, with ε_{Hs} over all points.

5.2 Cost

Cost relative to DIA.

Method	S_{nl} only	Single time step	10-day run
DIA	1	1	1
SWRT	3.7	1.7	1.6
SWRT Lite	1.4	1.14	1.12
WRT (vectorized)	66	16	14

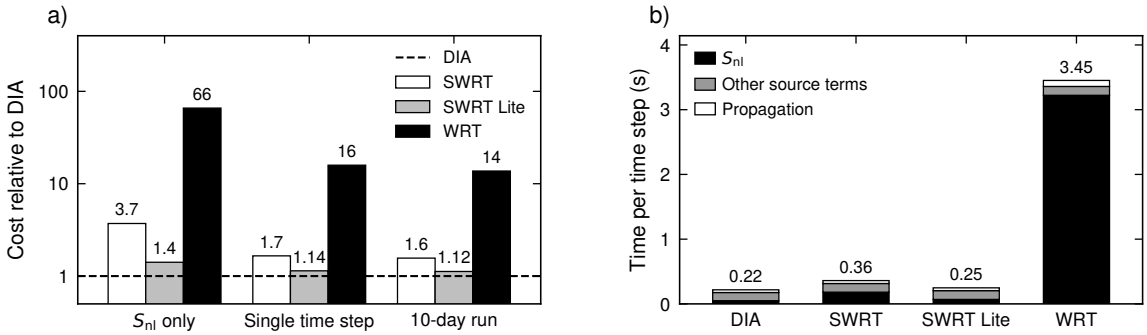


Fig. 3. Cost of S_{nl} in the global test. (a) Cost of SWRT (white), SWRT Lite (gray) and WRT (black) relative to DIA (dashed line) for S_{nl} only, a single time step and the 10-day run; (b) time per time step, split into S_{nl}, the other source terms and propagation.

5.3 Other spectral grids

Each file of section 3 run in the global test on its own spectral grid, against WRT on the same grid. Each cell gives SWRT / SWRT Lite / DIA, in %.

Directions × NFRE	First frequency (Hz)	ε_{Hs}	ε_{Tz}
36 × 36	0.0345	1.4 / 1.8 / 8.1	1.3 / 1.4 / 6.0
24 × 36	0.0345	0.5 / 2.2 / 10	0.6 / 2.0 / 7.0
24 × 30	0.0345	0.9 / 4.0 / 15	0.9 / 3.6 / 10
24 × 40	0.0345	0.5 / 3.4 / 9.5	0.6 / 2.1 / 7.0
12 × 36	0.0418	0.6 / 2.6 / 18	0.4 / 1.8 / 10
24 × 36	0.0418	1.7 / 2.1 / 9.7	1.6 / 2.0 / 6.9
36 × 36	0.055	0.7 / 1.1 / 7.2	0.7 / 0.9 / 5.7

5.4 Example spectra over ten days

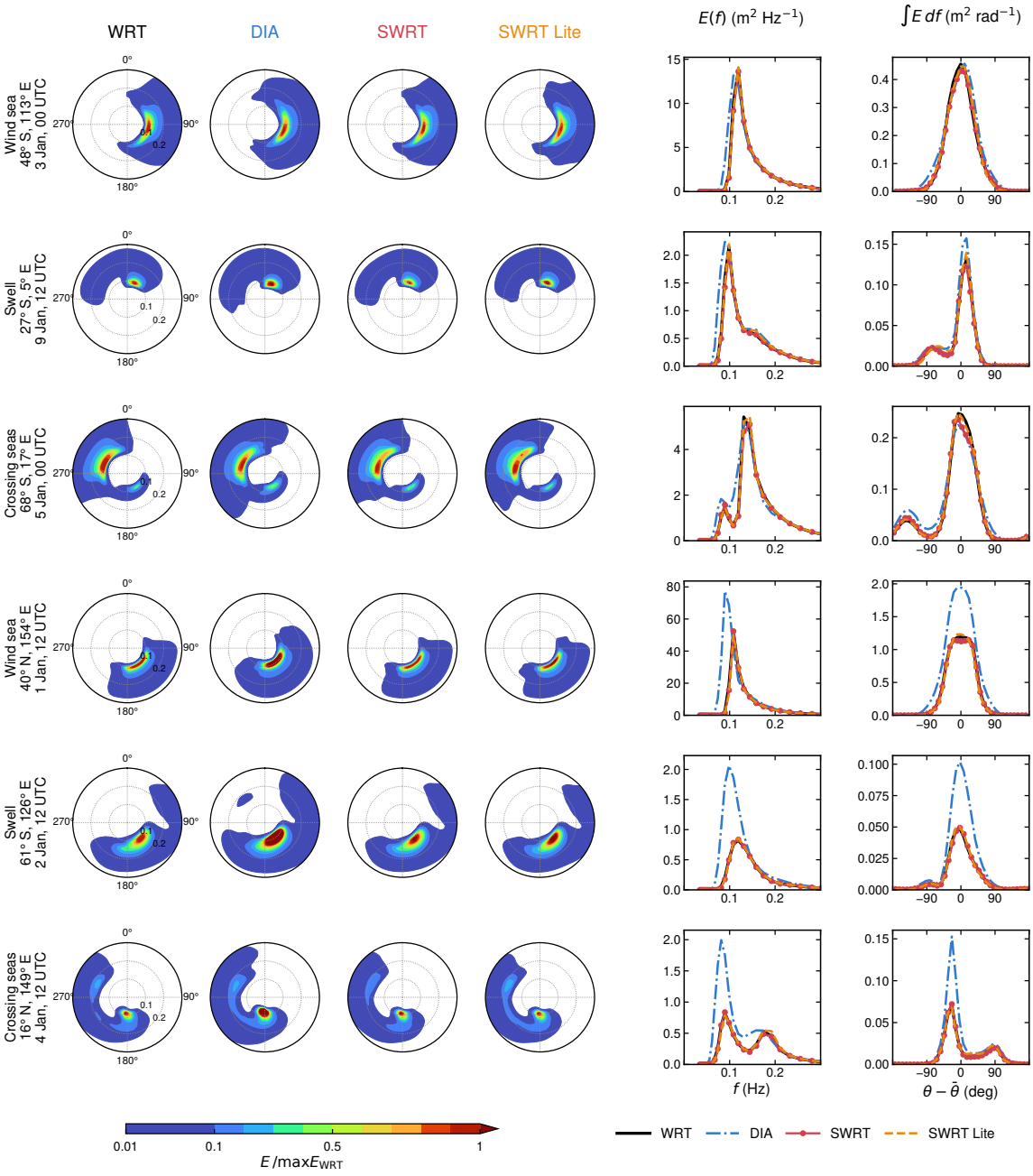


Fig. 4. Wave spectra of two wind seas, two swells and two crossing seas in the 10-day test. Left: E(f, θ) divided by the maximum of WRT; right: E(f) and the directional distribution ∫E df.

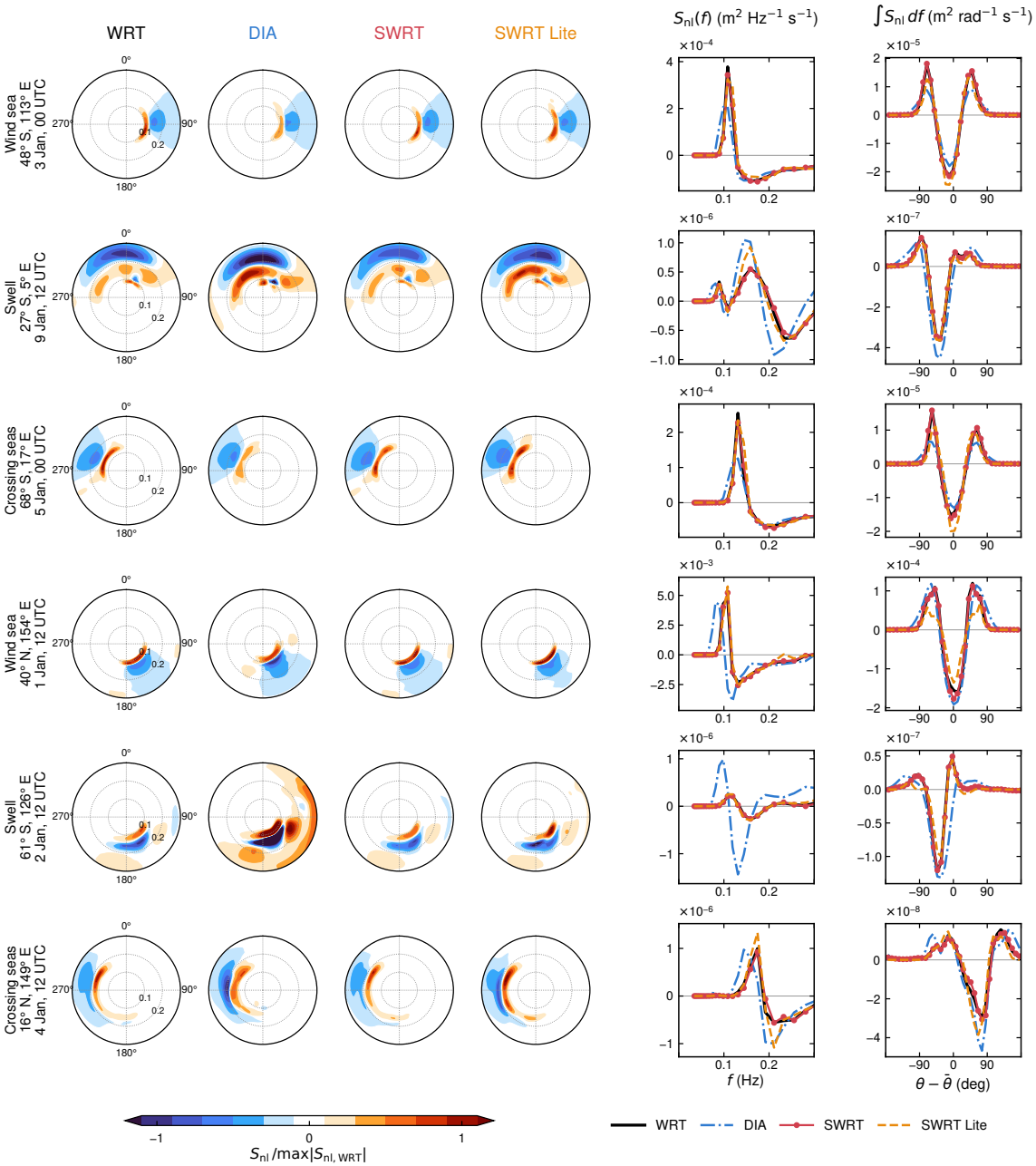


Fig. 5. S_{nl} of the spectra of Fig. 4. Left: $S_{nl}(f, \theta)$ divided by the largest $|S_{nl}|$ of WRT; right: $S_{nl}(f)$ and $\int S_{nl}\,df$.